Project Proposal & Literature Review

Blind Image Deblurring with a Compact Transformer

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Abstract

Motion blur degrades images captured while the camera or the scene is moving, and removing it from a single image without knowing the blur kernel (blind deblurring) is a hard inverse problem. We plan to study whether a small, parameter-budgeted Transformer or hybrid CNN–Transformer can deblur images competitively with a classical Wiener-filter baseline at low cost. We will use a subsampled version of the GoPro deblurring dataset as the main benchmark, plus a small synthetically blurred DIV2K subset as a sanity check for the Wiener baseline, where the blur kernel is known. Methods will be compared on identical test images using PSNR, SSIM, and parameter count.

1 Problem Statement & Motivation

A blurred image can be modeled as a sharp image convolved with a blur kernel (the point spread function, PSF), plus noise. If the PSF is known, classical deconvolution methods such as Wiener filtering or Lucy–Richardson (Richardson, 1972; Lucy, 1974) can restore the image. In real photographs, however, the kernel is unknown and often varies across the image, which makes the problem *blind*. Deep networks learn the mapping from blurred to sharp images directly from paired examples (Nah et al., 2017; Kupyn et al., 2019), but the best-performing models can be large (Kong et al., 2023; Mao et al., 2024). Our goal is to build a deliberately small model and measure how it compares to the classical baseline in accuracy and in parameter count.

This connects to the course material on image restoration and the Wiener-filter pipeline (the Wiener-filter pipeline from the Week 7 lab, as referenced in the project catalog).

2 Related Work

Classical deconvolution recovers a sharp image when the PSF is known (Richardson, 1972; Lucy, 1974); its weakness for real photographs is that the kernel must be supplied or estimated. Nah et al. (2017) addressed real dynamic-scene blur with a multi-scale convolutional network and released the GoPro pairs of blurred and sharp frames, which later work uses as a standard benchmark; the authors note that blur kernels are not provided because they are unknown. Kupyn et al. (2019) is the heavier learned reference point suggested by the project catalog.

Recent work applies Transformers to deblurring. Kong et al. (2023) observe that attention can be computed in the frequency domain with an element-wise product instead of a matrix multiplication, and use this in the decoder of an asymmetric encoder–decoder network. Mao et al. (2024) compute channel-wise self-attention inside local frequency-domain windows combined with an MLP gating

Table 1: Key references.

Reference	Year	Relevance to this project
Richardson (1972); Lucy (1974)	1972; 1974	Lucy–Richardson deconvolution: classical iterative baseline that assumes a known PSF.
Nah et al. (2017)	2017	Introduced the GoPro dynamic-scene deblurring dataset (our main benchmark).
Zamir et al. (2022)	2022	Efficient Transformer for high-resolution image restoration; a reference design for compact attention.
Kupyn et al. (2019)	2019	DeblurGAN-v2; heavier reference point named in the project catalog.
Kong et al. (2023)	2023	Frequency-domain Transformer for deblurring (recent, efficiency-oriented).
Mao et al. (2024)	2024	Local frequency-domain Transformer for deblurring (recent).
Liu et al. (2024)	2024	DeblurDiNAT: a compact encoder–decoder Transformer for deblurring; closest to our "compact" goal.

mechanism. Both build on the line of efficient restoration Transformers represented by Zamir et al. (2022). Liu et al. (2024) propose DeblurDiNAT, described by its authors as a compact encoder–decoder Transformer that alternates dilation factors in neighborhood attention to capture local and global blur, and that reports fewer parameters than its nearest competitors. These papers report strong accuracy; we have *not* yet measured any of these models ourselves, nor checked whether they fit our compute budget. We use them as reference points for design ideas and for the accuracy-versus-size trade-off, not as baselines we reproduce.

We do not claim a research gap beyond the project's stated objective: studying the accuracy-versus-parameter-budget trade-off of a small model against a classical baseline.

3 Dataset & Baseline Plan

Main dataset (GoPro). The GoPro dataset of Nah et al. (2017) contains 3214 blurred/sharp pairs: 2103 for training (22 sequences) and 1111 for testing (11 sequences). Because the official split is by sequence, train and test frames do not come from the same video. **Sanity-check dataset (DIV2K).** A small subset of sharp DIV2K images will be blurred synthetically with a known linear motion kernel, so that the Wiener filter can be checked in the setting where it is expected to work.

Subsampling plan. Uniformly take every k-th frame from each sequence, holding out whole sequences for validation so that validation frames are not near-duplicates of training frames. With $k = 5$ this gives about 420 training and 222 test pairs. The exact counts, selection method, and random seed will be recorded in the repository (`data/README.md`) and in the report.

Baseline. Wiener-filter deconvolution (course Week 7 pipeline); Lucy–Richardson is an optional second classical baseline. The Wiener filter needs a PSF. On DIV2K the PSF is known. On GoPro it is not, so we will use a single assumed linear-motion kernel whose length and angle are chosen on the validation set, and we will state clearly that this is a weak, non-blind setting that gives the baseline no per-image kernel estimation. The Wiener implementation follows the Week 7 lab

Table 2: Dataset summary.

Dataset	Subset used (planned)	Source / license
GoPro (GOPRO_Large)	Every 5th frame of each sequence: about 420 of the 2103 training pairs and about 222 of the 1111 test pairs; whole training sequences held out for validation; seed 42.	https://seungjunnah.github.io/Datasets/gopro. Released under CC BY 4.0 (stated on that page).
DIV2K	25 images sampled with seed 42 from DIV2K_valid_HR (100 images), blurred synthetically with a known linear motion kernel.	https://data.vision.ee.ethz.ch/cvl/DIV2K/. Academic research use only, as quoted in the Hugging Face DIV2K loader; the official page's terms were not independently checked.

pipeline.

4 Proposed Method

Planned design (tentative until the first experiments). The model takes a blurred RGB image and predicts the sharp image (or a residual added to the input). A small convolutional stem and encoder extract local features; a few Transformer blocks with an efficient (for example channel-wise) attention operate on the reduced-resolution features to capture global context; a convolutional decoder reconstructs the image. Compactness will be measured by total parameter count (and optionally inference time). Our design target is a budget of about 5 million parameters or fewer; this is a target we set ourselves, not a course requirement, and it will be revised after the first Colab test.

Evaluation protocol. PSNR and SSIM on the same held-out test images for every method, with identical preprocessing and a fixed random seed, plus parameter count. We make no claim that the proposed model will outperform the baseline, and we make no novelty claim from combining CNNs and Transformers; the final comparison will be reported as measured. A parameter-budget ablation is planned for the final milestone, as suggested by the project catalog.

5 Feasibility & Compute Plan

Compute tier: GPU-light. The team has no strong local GPU, so training will run on Google Colab (free tier or similar; the available GPU type is not guaranteed). The classical baseline runs on CPU. Time per training run has not been measured yet; our planning assumption is that each run fits in one Colab session, and we will measure a short run before Milestone 2. Subsampling (Section 3) keeps the data small enough for this setting.

5.1 Minimum Viable Deliverable (MVP)

Wiener baseline numbers (PSNR/SSIM) on the GoPro subset and the DIV2K sanity check, and one small model trained on the GoPro subset and evaluated on the same test images with PSNR, SSIM, and parameter count, giving one valid comparison table.

5.2 Stretch Goal

If the MVP is on track by Weeks 5–6: the parameter-budget ablation with several model sizes, Lucy–Richardson as a second baseline, and qualitative crops. A robotics-motivated extension (for example, the effect of deblurring on a downstream task) is not part of this proposal's scope; it can be reconsidered after the MVP is reached.

6 Risk Assessment

- **Unknown blur kernel for the Wiener baseline on GoPro.** Mitigation: DIV2K sanity check with known kernel; explicit statement of the assumed-kernel limitation.
- **Limited Colab compute and session time limits.** Mitigation: subsampled data, small model, checkpoints saved regularly, short runs first.
- **Team has limited deep-learning experience.** Mitigation: start from the smallest working model and verify each step with a short experiment before scaling up.
- **Dataset license or availability issues.** Mitigation: read the official license before use; fall back to the DIV2K synthetic variant if GoPro cannot be used.
- **Short milestone timeline.** Mitigation: the MVP is defined as the minimum valid comparison, and extras are optional.

7 Work Breakdown

The course template assumes three students; our team has two, so the three catalog roles (Data & Baseline, Model, Evaluation) are divided between two people. Each member owns a clear part of every milestone, and each reviews the other's code before it is merged.

Table 3: Milestone task allocation (draft; adapted from the project catalog entry).

Milestone	Zainab Ahmed	Rawda Abdelkhalek
M1 (Wk 3)	Literature review and related-work table; proposed-method design and parameter budget; Colab compute and timing test.	Dataset download and subset pipeline; Wiener baseline with PSNR/SSIM on the subset and the DIV2K sanity check; dataset and baseline sections.
M2 (Wk 6)	Compact model implementation, training loop, and first trained checkpoint on the GoPro subset.	Full Wiener baseline on the GoPro subset; evaluation scripts (PSNR, SSIM, parameter count); first comparison table.
M3 (Wk 9)	Final model training and the parameter-budget ablation; architecture figure.	Final comparison on the shared test images; qualitative result figures; demo video and slide deck assembly.

Report sections are written by the member who owns the corresponding work, and both members prepare and present the final presentation.

AI-assistance disclosure. Code scaffolding and drafting support for this proposal were AI-assisted (Claude); all references were checked and all results will be run and verified by the team.

References

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